



PYTHON

ORIENTATION



IN THIS PRESENTATION

- Full Stack Overview
- What is Python?
- Interpreter vs Compiler
- Python Syntax

FULL STACK OVERVIEW

- Full Stack refers to the entire depth of a computer system application, and full stack developers handle both the front end and the back end
- The front end includes everything a client can see and interact with
 - Front-end devs work to optimize end-user visible parts of an application
 - This typically involves creating web pages and mapping user interactions to work with the application
- The back end refers to all the servers, databases, and other internal architecture that drives the application
 - Back-end developers work on the code that communicates with servers, databases, and other applications and conveys that information to the front-end interfaces

WHAT IS PYTHON?

- Python is a programming language that comes with a library of data and an interpreter that reads and executes the code we create
- Python got its name from Monty Python because the creator, Guido van Rossum, was a fan of the show and wanted the language to be fun
- It is used for its flexibility and can be used for simple scripting or creating complex enterprise applications
- Why Python?
 - Simplicity
 - Open-sourced
 - High-level language
 - Interpreted
 - OOP and functional
 - Portable
 - Many 3rd-party libraries that integrate with the base library

INTERPRETER VS COMPILER

- A compiler is software that takes the entirety of your project source code and converts it into a form that can be executed by your operating system
 - Usually a binary(.bin) or executable(.exe) file
 - Optimizations can be applied during compilation
 - Can detect issues with your code before execution
 - C++, Java, Go
- An Interpreter is similar to a compiler in that it converts source code into machine code, but it does it line by line, not all at once
 - Interpreters are specific to their operating system, not the application
 - Interpreted languages are typically easier to debug
 - Less optimized than their compiled counterparts; not used when high levels of performance are needed
 - Python, Ruby, JavaScript

PYTHON SYNTAX

- Python's functional syntax is more simplified than some other OOP languages
 - Java: `int age = 15; System.out.println(age);`
 - Python: `age = 15 print(age)`
- Python does not require semicolons, type declaration, or braces
- Instead, our code is very short and readable and uses indentations in place of braces
- Python naming conventions: Screaming Snake Case, Snake Case, and Pascal Case
 - Screaming Snake Case: All applicable characters must be uppercase and have underscores between words
 - Used for constants
 - `MY_VARIABLE_NAME`
 - Snake Case: must start with a lowercase letter and have underscores between words
 - Used for classes
 - `my_variable_name`
 - Pascal case: must start with an uppercase letter and all words start with an uppercase letter
 - Used for everything else
 - `MyVariableName`

PYTHON SETUP

- For this course, we will be using VS Code: [Download Visual Studio Code - Mac, Linux, Windows](#)
- [Welcome to Python.org](#)
- Ensure that you have Python installed:

- Open CMD window

Check if Python is installed

```
python --version
```

|

macOS

```
brew install python3
```

|

Ubuntu/Debian

```
sudo apt update && sudo apt install python3
```

|

Windows

```
winget install Python.Python.3.12
```